

TP Infovis 2025: Healthcare Visual analytics Tool

This TP focuses on designing an interactive healthcare visual analytics tool using synthetic healthcare data.

Objectives

- Data preprocessing and feature engineering
- Interactive visualization using D3.js
- Geospatial visualization using ArcGIS JavaScript API
- Conceptual understanding of multi-class classification

Work Organization

- Group size: Binôme (max 3 students)
- Duration: 2 weeks (before the last exam)
- Technologies: HTML, CSS, JavaScript, D3.js, ArcGIS JS

Dataset

<https://www.kaggle.com/datasets/prasad22/healthcare-dataset/data>

Target variable (classification):

- Test Results $\in \{ \text{Normal, Abnormal, Inconclusive} \}$

Main attributes include:

- Patient info: Age, Gender, Blood Type
- Medical info: Medical Condition, Medication, Test Results
- Administrative info: Hospital, Doctor, Admission Type, Insurance Provider
- Financial info: Billing Amount
- Temporal info: Date of Admission, Discharge Date

Part 1 – Data Preparation

Students must:

1. Load the CSV file using JavaScript.
2. Handle:
 - Missing values (if any)
 - Data type conversion (dates, numbers)
3. Create derived features such as:
 - Length of stay = Discharge Date – Admission Date
 - Age groups (e.g., 0–18, 19–40, 41–65, 65+)

Part 2 – D3.js Vis

Design an **interactive visualization** with **at least 4 visualizations**, including:

Proposed Visualizations should be for:

1. Distribution of Test Results

2. Medical Conditions vs Test Results
3. Billing Amount Analysis
4. Patient Demographics
5. Interactivity (Required)

Part 3 – Geospatial Visualization

Using **ArcGIS JavaScript API**, students must:

1. Display hospitals on a map (coordinates can be:
 - Provided manually
 - Approximated by city/region
2. Each hospital marker should show:
 - Hospital name
 - Number of patients
 - Average billing amount
 - Dominant test result category
3. Enable interaction:
 - Clicking a hospital updates D3 charts
 - Hover popup with summary statistics

Part 4 – Analytical Perspective

Without implementing a full ML model, students must:

1. Treat Test Results as a **multi-class classification problem**
2. Identify:
 - Most influential features (Age, Condition, Admission Type, etc.)
 - Potential biases or limitations
3. Propose:
 - Which model could be used (e.g., Random Forest, XGBoost, Neural Network)
 - Which features would be selected and why

Deliverables

- Web application files
- Report (PDF)

Evaluation Criteria

- Data preprocessing: 20%
- D3.js visualizations: 30%
- ArcGIS integration: 20%
- Analysis: 15%
- Code quality & UI: 15%

Inspiration (Allowed References)

Students may use the following **for inspiration only** (no copy-paste):

- Healthcare dashboard discussion
<https://www.kaggle.com/datasets/prasad22/healthcare-dataset/discussion/609659>
- Medium article (QlikSense visualization)
<https://medium.com/@molly-nakalyango/healthcare-dataset-analysis-visualization-qliksense-application-de276912608d>

Kaggle notebooks or ready-made code **must not be reused**.

Good luck